

Energy-Guided Sampling for Controllable Text Generation

From Langevin Dynamics to Amortized Inference

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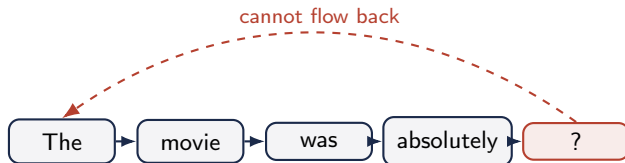
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Thesis Defense

The problem autoregressive models cannot solve natively

- Autoregressive generation: one token at a time, left to right.
- A token is fixed the moment it is emitted.
No mechanism to revise an earlier token in light of a later one.
- Global properties (sentiment, factual constraint) live in the *whole* sequence.



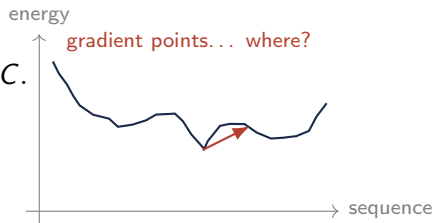
Control means steering the *completed* object. That is a global problem.

The energy-based premise, and its one assumption

Treat the frozen likelihood as an **energy** over sequences and sample from it:

$$p(\mathbf{x}) \propto \exp(-E(\mathbf{x})), \quad E = -\log p_{\text{LM}} + \lambda C.$$

- Add any constraint C at inference time.
- One frozen model, any objective.
Plug and play.



A jagged landscape: the local slope need not lead anywhere good.

The untested assumption:

the gradient of the frozen likelihood is a usable search direction on discrete text.

The machinery: Langevin dynamics

Sample $p(\mathbf{s}) \propto e^{-E(\mathbf{s})}$ with a gradient-guided chain:

$$\mathbf{s}_{t+1} = \mathbf{s}_t + \frac{\epsilon_t}{2} \nabla_{\mathbf{s}} \log p(\mathbf{s}_t) + \sqrt{\epsilon_t} \boldsymbol{\xi}_t$$

- **Drift:** uphill.
- **Diffusion:** a *sample*, not a mode.
- Needs a **Lipschitz** drift.

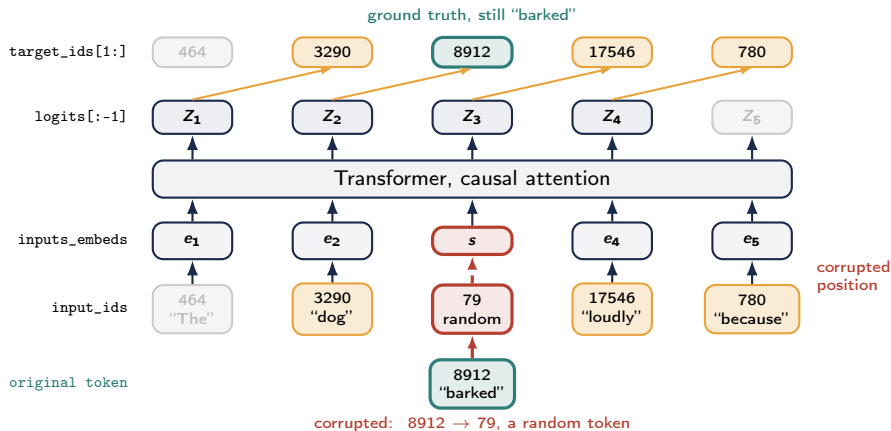
small steps, smooth surface



Text is *discrete*: there is no small step.

Every method here forces this update onto discrete tokens.

Scoring a sequence: two arrays, offset by one



$$E(\mathbf{x}) = -\sum_t \log \text{softmax}(\mathbf{Z}_t)[\text{id}_{t+1}]$$

The target row is the input row, shifted by one. The same sequence is present as continuous vectors and as integers.

Research questions

RQ1 Does the *input-embedding* gradient give a usable proposal direction? If not, what does?

RQ2 What does the Metropolis–Hastings correction do in each sampler?

RQ3a Does a GFlowNet on this reward learn a high-reward policy?

RQ3b Does its tuned energy become more navigable?

E (extension) Can an additive constraint steer generation here?

Two ways out, if the gradient fails. Repair the search (RQ1, RQ2), or abandon the search and *train a policy* to absorb it (RQ3). The talk takes them in that order.

Three candidate explanations

Suppose the gradient does not help. *Three* things could be at fault.

	The claim	Decided by
A target	No local search could profit from this likelihood. Then the proposal is beside the point.	search the same energy <i>without</i> a gradient
B model	Teacher forcing only ever asks for the next token, never to <i>rank a substitution</i> . Then only retraining repairs it.	re-differentiate the <i>same</i> frozen weights
C coordinates	The information is there and this derivative discards it. A proposal reading the <i>output side</i> recovers it; one seeing the <i>right context</i> recovers more.	a ladder of proposals in one exact chain

Everything that follows is that elimination, in that order.

The setup: one task, five energies, one metric

Task. Masked-token recovery: corrupt tokens, ask the sampler to put them back.

200 sequences per configuration, paired seeds across arms.

Metric. Mean KL to the ground-truth fill. Floor of 0.

Five energies.

- **GPT-2 Large**, fine-tuned. The reference.
- **Three GFlowNet-tuned** variants of it.
- **Llama-3 8B**. Cross-architecture.

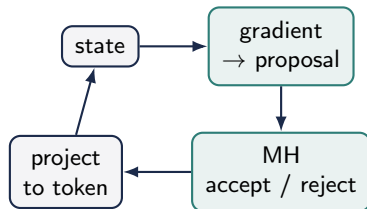
145 configurations. Equivalence margin fixed in advance at 0.327.

Two samplers, implemented faithfully

- **DLS** (discrete): stays in token space.
- **CLS** (continuous): relaxes into embedding space, projects back.
- **Metropolis–Hastings** correction, applied exactly.
- Step size by **oracle calibration**.

DLS scores a candidate v against x_i , with $\mathbf{g} = \nabla_{\mathbf{e}_i} \log p_{\text{LM}}$:

$$\log q(v \mid x_i) \propto \underbrace{\frac{1}{2} \mathbf{g}^\top (\mathbf{e}(v) - \mathbf{e}(x_i))}_{\text{gradient alignment}} - \underbrace{\frac{1}{2\alpha} \|\mathbf{e}(v) - \mathbf{e}(x_i)\|^2}_{\text{distance}}$$



the step the field skips

A fixed annealed schedule, never stopped early.

This $\mathbf{g}$, the input-embedding gradient, is the object under test.

First: what is the proposal we are about to ablate?

Entropy measures how spread out a distribution is, in *nats*.

- Concentrated on one token: 0.
- Flat over all 50,257:
 $\log |V| = 10.825$, the ceiling.

So entropy answers: *how many tokens is the proposal really choosing between?*

The calibrated proposal, measured

at its *sharpest*: 10.8248 nats

uniform ceiling: 10.8249 nats

Effectively choosing among $\approx 29,000$ tokens at every step.

The gradient term spreads the logits by ≈ 0.009 nats. The softmax that consumes them has a 10.82-nat budget, so the gradient moves the proposal by about one part in a thousand.

The calibrated proposal is numerically uniform.

So a 5×5 sweep over step size and temperature re-tests everything below where the gradient *does* shape the proposal.

The central result: is the *direction* worth anything?

The control. Take the true gradient, throw away its direction, keep its length exactly. That is the *norm-matched random* arm. The two differ in direction and in nothing else.

Read the whole chain, not its end.

Three summaries of the same run:

- **last step**: where it stopped.
- **chain mean**: its typical state.
- **best state**: the best it ever reached.

Same picture on all five energies, and in the sweep recovery never exceeds 2% in any of the 25 cells.

Policy minus norm-matched random, $n = 1000$ paired

last step	+0.133	$[-0.063, +0.326]$
chain mean	+0.136	$[-0.012, +0.287]$
best state	+0.033	$[-0.079, +0.145]$

Margin ± 0.327 nats, written down *before* the test: any gap smaller than this is too small to matter. All three intervals fit inside it.

So this is *certified equivalence*, not “we failed to find a difference”.

Hypothesis A: maybe the energy itself is hopeless?

Then try something simpler: search the *same* energy without ever differentiating it.

Method (no gradient of the energy)	KL	exact
Ground-truth fill (floor)	0.00	—
Corrupted token left in place	9.14	0%
Best Langevin cell (<i>uses the gradient</i>)	~ 6.4	0.0%
Gradient-free Gibbs sampler	6.69	18.5%
Top- k rescore, one forward pass	4.43	33%

Gibbs: blank one position, ask the model for its distribution there, draw a token, move on. Repeat. No gradient anywhere.

A is rejected: the energy is searchable, just not by this derivative.

And the cheap methods are the good ones: no backward pass at all.

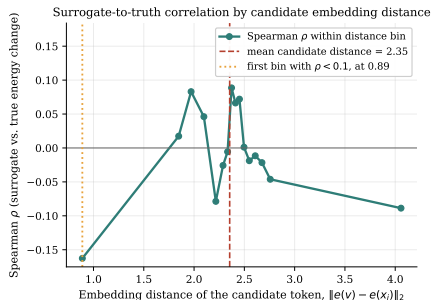
Why, part 1: the approximation dies before the first step

The proposal ranks a candidate by a straight-line estimate of what swapping it in would do:

$$\hat{\Delta}(v) = \mathbf{g}^\top (\mathbf{e}(v) - \mathbf{e}(x_i))$$

A straight line only fits a curve *near* the point it is drawn at. So: how far does it stay accurate?

Measured: correlate the estimate against the truth, *within bins of distance*.

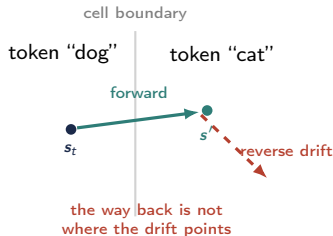


Correlation with the true change, by how far the swap moves you.

It is at zero by distance 1.82, which is the average gap between *neighbouring* tokens: the smallest move that changes any word at all.

Every legal move lies outside the range where the estimate works.

RQ2: what does the accept/reject step actually do?



Cross a boundary and the projected token changes, so the drift on the far side is computed for a *different* prediction. The reverse move is not proposable.

The acceptance ratio is a product of two things:

$$\alpha = \underbrace{\frac{\pi(s')}{\pi(s_t)}}_{\text{is it better?}} \cdot \underbrace{\frac{q(s_t | s')}{q(s' | s_t)}}_{\text{could we come back?}}$$

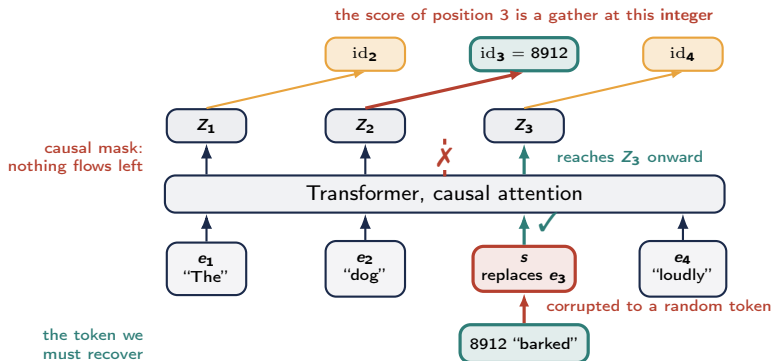
Boundary-crossing moves, measured

$$\begin{aligned} \text{"is it better?"} &= e^{+4.60} \text{ yes} \\ \text{"could we come back?"} &= e^{-1325} \end{aligned}$$

The second term is 10^{575} times smaller. Good moves are rejected for being *unreturnable*, not for being bad.

Continuous: disabled (3.7–8.6%). Discrete: rescued (100% within-cell).

Back to the two arrays: what the gradient cannot reach



Z_2 was built from e_1, e_2 alone, and id_3 is the *integer* s stands in for. So that derivative is exactly 0.

$$\boxed{\text{future term, what it had } g^{\top}(e(v) - e(x_i))} + \boxed{\text{self term, read off the logits } \log p(v \mid x_{<i})}$$

Hypothesis B: same model, same sampler, different derivative

Relax the token indicator $\mathbf{z}_i$ in *both* of its roles:

$$\frac{\partial \tilde{L}}{\partial \mathbf{z}_i[v]} = \underbrace{\log p(v \mid \mathbf{x}_{<i})}_{\text{self, discarded}} + \underbrace{\mathbf{g}^\top \mathbf{e}(v)}_{\text{future}}$$

Same weights, same sampler, same energy. Only the derivative changes.

ρ (near): how well the estimate *ranks* candidates against the truth, over the swaps that are actually reachable. 1 = perfect order, 0 = no order.

Surrogate	ρ (near)	exact
input embedding	0.03	0.0%
token indicator	0.73	40.0%

Llama-3 8B reaches 41.0% where every input-embedding run gives 0.0%.

$$0 \rightarrow 40\%$$

B rejected in its strong form

A clean confirmation: at the last position the gradient is provably zero

The final token's embedding feeds no prediction that exists, so

$$\nabla_{\mathbf{e}_L} \log p_{\text{LM}}(\mathbf{x}) = 0 \quad (\text{exactly}),$$

while the energy *difference* between candidate final tokens is exactly the model's conditional log-ratio.

Energy maximally usable; gradient provably useless.

Position	LM grad norm
final token	0.0000
second-to-last	> 0
middle	> 0

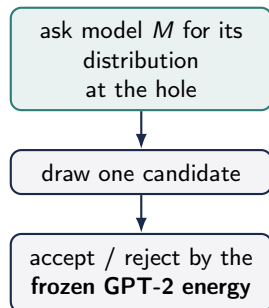
Langevin recovers 0% at all three;
exact-energy methods 18–55%.

Hypothesis C: swap the proposal, keep everything else

An autoregressive model can only ever look *left*. To fill a hole it has no path from the words *after* it.

Two model families can look both ways:

- **SEDD**, a discrete diffusion model: re-mask a position, re-denoise it from both sides.
- **RoBERTa**, an ordinary masked LM: the same query, and its native training task.



An independence sampler: M only *suggests*. The energy being sampled is still GPT-2's, unchanged. Same 200 sequences, same corruptions, same metric.

Only the box marked M changes. So whatever moves is caused by the proposal.

The conditioning ladder

Proposal	can see	exact
input-emb. gradient	nothing	0.0%
norm-matched random	nothing	0.0%
uniform draw	nothing	0.5%
GPT-2's own conditional	left only	23.5%
SEDD (score-trained)	both sides	39.0%
RoBERTa MLM (no score obj.)	both sides	44.5%

The uniform draw reproduces the flagship Langevin sampler, KL 6.538 vs 6.541, from separate code. So the pipeline leaks nothing.

Two claims, both confirmed

Output side beats no side:

0.5 $\rightarrow$ 23.5.

Both sides beat left only:

23.5 $\rightarrow$ 44.5.

Gradient-free comparator on the same task: top- k rescore 33%. So 44.5 is a gain over 33, not over nothing.

What orders this is access to context, not the training objective.

GFlowNet amortization does not escape the failure

A GFlowNet only ever *evaluates* the reward, never differentiates it.

- Three failure modes: **capacity collapse**, **length collapse**, **reward hacking**.
- Tuning moved the energy by 31–57 nats. . .
- . . . yet Langevin sampling on the tuned energy is *indistinguishable* from the untuned base.

Changing the model through this objective did not make the local surrogate useful.

The extension: can we actually steer it?

The original promise: $E = -\log p_{\text{LM}} + \lambda C$, with C a sentiment classifier. Does adding C move the output?

Every number below is a contrast, never a raw gain:

steering = %target hit(constraint only) – %target hit(randomized constraint)

Same constraint, same magnitude, *direction destroyed*. Raw gains are bias-dominated, so only the subtraction isolates the *direction*.

Sampler	Target	cons_only – cons_random
MuCoLa-style continuous	negative	+27.3
MuCoLa-style continuous	positive	+36.7
Discrete sampler (ours)	negative	0.0
Discrete sampler (ours)	positive	+0.7

Points. The constraint gradient's direction carries signal where the fluency gradient's never did. But the arm that steers has no fluency term holding it down.



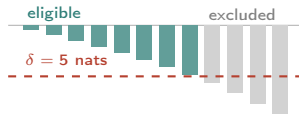
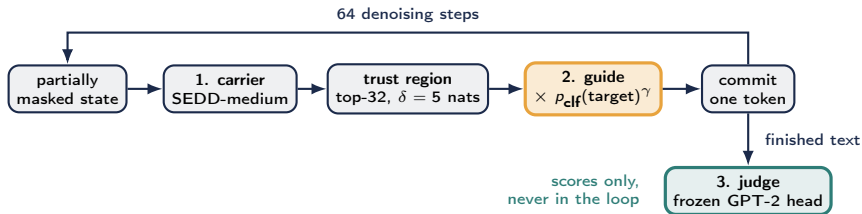
Grey: the shared sentiment drift, present even with no constraint at all, and it cancels.

Teal: what the direction adds.

A constraint added to a landscape no sampler can navigate is added to nothing.

So put the constraint on a landscape that *is* navigable

Carry the constraint on the bidirectional proposal instead. Three instruments, strictly separated:



candidates, by log-prob below the carrier's own top choice

- Steering works *on the instrument it optimizes*: +9.3 to +25.7 pts, both labels, every γ .
- Transfer to the judge is bounded by how far the two instruments *agree*: 56 $\rightarrow$ 72 $\rightarrow$ 80% as the text moves on-domain.
- The trust region keeps 28.4 of 32 candidates eligible and removes the fluency cost (7.1 $\rightarrow$ 11.0 nats without it).

Control follows navigation. Explored, not settled: a repair on *another* landscape.

Takeaways: the four questions, answered

RQ1 direction	No. Same length, random direction: indistinguishable, inside a margin fixed in advance, on 1000 paired sequences. The same frozen weights differentiated at the token's <i>identity</i> : 0% → 40%.
RQ2 correction	Opposite things in the two samplers. In the discrete one it supplies all of the selection pressure. In the continuous one it disables the sampler: a good move ($e^{+4.60}$) is rejected because the way back is unproposable (e^{-1325}).
RQ3 amortization	No escape. The policy reaches high reward by exploiting it, and tuning moved the energy by 31 to 57 nats without making the local surrogate any more useful.
E control	Only where the proposal can move. Nothing on the frozen autoregressive landscape; on a bidirectional one the target moves, bounded by how far the guiding and judging instruments agree.

One thread through all four: what a sampler can *condition on* and *read*, not how hard it searches.

What it comes down to

The signal was in the model all along.

The derivative threw it away.

- Relax the token *indicator* instead of the input embedding, or read the proposal off the model's output side. Same frozen weights, 0% $\rightarrow$ 40%.
- Let the proposal see the right context as well, and it goes further: RoBERTa 44.5%, never trained on scores.

Scope: the null is about an input-embedding gradient of a likelihood whose target token enters as a discrete index. MuCoLa and COLD keep that self term, and are not refuted by it.

Thank you.

Questions?

Post-submission: MuCoLa's energy in my own code

Identical code, optimizer, projection, initialization, step count and constraint weight. The *only* difference is how the target token enters the language-model term.

mucola_faithful

"Once upon a time, there was a king who ruled over a large land. His subjects were very happy, and he"

cls_thesis (my energy)

"Once upon a time environmentalCNN Unicornrosso PE ##YD vitality ped Eh Karn blades..."

	mucola_faithful	cls_thesis
mean final LM log-prob	-38.5	-353.9
gradient at last position	$\mathbf{h}_{n-1}$: norm 50.5592, cosine 1.000000	exactly 0.000e+00
steering-head "accuracy"	40.0%	53.3%

- Verified first: the energy at a real token sequence matches the ordinary log-likelihood to 0.000e+00.
- MuCoLa's energy writes fluent English; the energy I implemented writes word salad, and the final-position identity says exactly why: 50.56 against exactly zero.
- Note the trap in the last row. The blind energy scores *higher* on the steering classifier while producing gibberish. Hence every adherence number here is paired with an external fluency number, and the classifier that steers is never the one that scores.

Backup: the self gain

self gain (pts) = %guided hits the target label – %unguided hits it, both scored by the guiding classifier

On-domain trust-region cells, $n = 300$ prompt-paired generations each; unguided and guided share a seed per prompt.

γ	target	unguided	guided	self gain [CI]	judge gain [CI]
2	negative	48.0	66.0	+18.0 [12.3, 24.0]	-2.3 [-7.0, 2.3]
2	positive	52.0	61.3	+9.3 [3.7, 15.0]	+6.7 [1.7, 11.7]
4	negative	48.0	73.7	+25.7 [19.7, 31.7]	-0.7 [-5.3, 4.0]
4	positive	52.0	69.7	+17.7 [12.0, 23.3]	+7.3 [2.7, 12.0]

- The guide's own verdict. It moves at every strength and for both labels, so the mechanism does what it was built to do.
- The judge is a different instrument, and the two agree on 56% off-domain, 72% on on-domain generations, 80% on real sentences. The judge column tracks that agreement.
- The unguided baselines differ for the same reason: 53.0 / 47.0 read by the judge, 48.0 / 52.0 read by the guide.

Backup: the two classifiers

	guiding classifier (the guide)	judge (independent)
reads	token ids <i>including</i> MASK: the partially masked state mid-denoising	the frozen GPT-2 Large last hidden states of finished text
body	own embedding table over 50,258 ids (50,257 + MASK), mean-pool, Linear-Tanh-Linear, noise level appended as an input feature	mean-pooled linear probe on the frozen backbone: Linear(1280 $\rightarrow$ 256)-Tanh-Linear(256 $\rightarrow$ 2)
trained on	SST-2, each example corrupted by <i>SEDD's own</i> forward absorbing process, $t \sim U(0,1)$, each token $\rightarrow$ MASK with probability $(1 - \epsilon)t$	SST-2, clean text, backbone frozen throughout
accuracy	81% held out at low noise; 79.7% on the real held-out sentences	88% on those same sentences
role	guides only: reweights the carrier's top-32 at every commitment	grades only: never inside the loop

- GPT-2 cannot ingest the MASK id: 50257 sits outside its embedding table. A classifier that reads mid-denoising states therefore needs its own vocabulary, so the separation is forced by the architecture, and then kept on purpose.
- The guide is trained on exactly the states it is later queried on.

Backup: SEDD and RoBERTa, where the proposal is read and where it enters

Where the proposal is read off

- **SEDD** (Lou et al. 2024), absorbing graph, dim $50,258 = 50,257 + \text{MASK}$, GPT-2 tokenizer so no retokenization. The raw output $f_\theta(\mathbf{x}, \sigma)$ is *already* a log-space score.
- Read-out: set position m to MASK, one forward at $\sigma = 0.1$, drop the MASK column, log-softmax over the real vocabulary $\Rightarrow \log q(v)$.
- **RoBERTa-large**, an ordinary MLM, never score-trained: put <mask> at m , one forward, log-softmax of the MLM head.
- Tokenizer bridge: RoBERTa id $\rightarrow$ GPT-2 id for every token whose surface re-encodes to exactly one GPT-2 token, 99.95% coverage, restricted and renormalized.

Where it enters the sampler

the proposal logit $t_2[v]$, and nothing else

$$\frac{1}{2} \mathbf{g}^\top (\mathbf{e}(v) - \mathbf{s})$$

$\Downarrow$ the one thing swapped

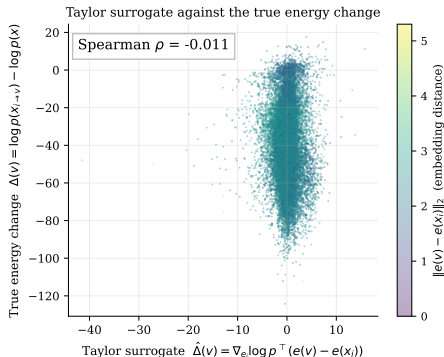
$$\frac{1}{2} \log q(v \mid \mathbf{x}_{\setminus m})$$

- The accept step is *untouched*:
 $\alpha = \min\{1, \frac{p(\mathbf{x}') q(\mathbf{x})}{p(\mathbf{x}) q(\mathbf{x}')} \}$ with p the exact frozen GPT-2 Large sequence likelihood.
- q does not depend on the current token, so this is an *independence sampler* and the Hastings ratio is closed form.
- Same sequences, same per-index corruption seeds, same energy, same KL metric. One variable moves.

Exact recovery: gradient 0.0 $\rightarrow$ left-conditional 23.5 $\rightarrow$ SEDD 38.5/39.0 $\rightarrow$ RoBERTa 44.5; uniform through the same code 0.5.

Backup: the linearization radius and self-term blindness

- The surrogate correlates with the *future* term ($\rho = 0.03$), never the *self* term ($\rho = -0.10$).
- Mean absolute self term 15.0 nats; future term 24.2 nats.
- The input-embedding gradient is structurally blind to the candidate's own left-context fit, which enters as a discrete index.



Backup: the Metropolis–Hastings chain

$$\alpha = \min \left\{ 1, \frac{p(\mathbf{s}') q(\mathbf{s} \mid \mathbf{s}')}{p(\mathbf{s}) q(\mathbf{s}' \mid \mathbf{s})} \right\}$$

- Target ratio: prefers good states. Proposal ratio: enforces reversibility.
- MALA needs a Lipschitz drift (Roberts and Tweedie 1996; Dalalyan 2017; Durmus and Moulines 2017).
- The projected target is piecewise constant; the differentiable pathway's drift jumps across a cell boundary, so the reverse density vanishes.

Backup: what the sampler does without the correction

- Correction off: a uniform random walk that never localizes. Mean KL *rises* from 8.765 to 9.499; the chain still changes its token on 99.1% of final-decile steps.
- So the late entropy dip is the annealing floor, not a quench: no freezing, no convergence to a local optimum.
- Correction on: the same uniform proposal becomes an independence sampler, the accept/reject filter does all the work, final KL 6.541 at a 9.98% acceptance rate.

Backup: likelihood is a poor objective, yet used everywhere

- The **likelihood trap**: the lowest-energy text is the most degenerate (Holtzman 2020, reproduced here). Within-strategy repetition correlation up to +0.91.
- Resolution: likelihood ranks single-token substitutions well (used by the KL metric and Gibbs), but maximizing absolute sequence likelihood in free generation is degenerate.
- So using it as the KL metric, the reward, and the SEDD comparison is consistent: relative-to-reference fit, not absolute likelihood as quality.

Backup: MuCoLa / COLD, and why they appear to work

- The continuous sampler follows their state *geometry* faithfully, a continuous embedding state with a projection after every update, and adds the correction those works omit. It does *not* follow their *energy*: see backup 33.
- Without the exact correction the update, *as reimplemented and measured here*, behaves as an early-stopped biased optimizer rather than a sampler from the stated energy. That is a statement about the mechanism, not a reinterpretation of those systems' published results, whose tasks, tuning and evaluation this study did not reproduce.
- The constrained *mucola* arm is that comparison, run at the energy level (final KL under the same base model), not the task level.

Backup: task design and corpus

- Masked-token recovery on WikiText-2 validation; the base GPT-2 Large is fine-tuned on ROCStories (out-of-domain test).
- In-domain ROCStories diagnostics reproduce the null and the likelihood trap: not a domain-shift artifact.
- The KL metric compares the recovered token's next-token conditional against the ground-truth-conditioned reference; synonyms score well.
- Llama-3 is a cross-architecture control, never on a shared numerical axis. One Llama contrast is nominally significant against the gradient, but it is one of dozens uncorrected, is measured with the correction off, and does not reproduce on GPT-2 sharp cells; the supported claim is *indifference*, not anti-guidance.

Backup: why did the temperature flatten everything?

- Proposal logit =
$$\left[\underbrace{\frac{1}{2} \mathbf{g}^\top (\mathbf{e}(v) - \mathbf{e}(x_i))}_{t_2} - \underbrace{\frac{1}{2\alpha} \|\mathbf{e}(v) - \mathbf{e}(x_i)\|^2}_{t_1} \right] / T.$$
- At the calibrated cell $T = 5$ divides a spread of a few nats into a 10.82-nat entropy budget: the softmax is uniform whatever the surrogate says.
- Sharpness from the *distance* term is sharp about *staying put* ($t_2/t_1 = 1.8$); sharpness from the *surrogate* is sharp about the ranking ($t_2/t_1 = 229$). Only the second tests the direction.
- The recovering cell needs large ϵ (kills t_1) and low T together.

Backup: what exactly is the token-indicator derivative?

Relax the token indicator $\mathbf{z}_i$ from its simplex vertex, in *both* roles:

$$\tilde{L}(\mathbf{z}_i) = \sum_v \mathbf{z}_i[v] \log p(v \mid \mathbf{x}_{<i}) + L_{\text{future}}(\mathbf{E}^\top \mathbf{z}_i)$$

- First term: the current position's own log-likelihood with the discrete *target* relaxed. Second term: the suffix, with the relaxed indicator mapped through $\mathbf{E}$ to an input embedding.
- At a vertex $\tilde{L}$ equals the sequence log-likelihood, so this is a relaxation of the same object.
- It is *not* $\mathbf{E}^\top \mathbf{g}$, the ordinary one-hot input gradient through the embedding lookup: that carries the future term only. The whole difference is the self term.

Backup: why does RoBERTa beat SEDD?

- Both are bidirectional; only SEDD is score-trained. RoBERTa 44.5% vs SEDD-medium 39.0%, so the score objective is not what buys recovery.
- RoBERTa-large is bigger and trained on more text than SEDD-small/medium, and masked-token prediction *is* its native query at any noise level, whereas SEDD answers it at a schedule point.
- So the claim is an *ordering by conditioning*, not an isolation of it and not a ranking of the two models: the arms differ in scale, corpus and architecture as well, and only the tokenizer and the chain are held fixed across all of them. The SEDD arms are pilots.
- Tokenizer bridge: 99.95% of RoBERTa ids map to one GPT-2 token; no ground-truth token fell outside it. Uniform control through the same code: 0.5%, so the pipeline leaks nothing.

Backup: why a GFlowNet, and the task-comparability caveat

- A GFlowNet samples in proportion to reward (not the mode), and never differentiates it, so a gradient pathology would be invisible to it.
- It reformulates infilling as left-to-right generation under a restructured prompt; the samplers do in-place bidirectional substitution.
- The comparison is therefore run at the energy level, not the task level; the left-to-right reformulation is not adopted for the samplers because it would dissolve the revision capability under study.

Backup: how this differs from MuCoLa and COLD

Method	how the token's <i>own</i> score enters	self term
MuCoLa	softmax numerator $h_n^\top \tilde{\mathbf{e}}_{n+1}$, target relaxed	kept
COLD	$\sum_v p_{\text{LM}}(v \mid \tilde{\mathbf{y}}_{<t}) \log \text{softmax}(\tilde{\mathbf{y}}_t(v))$	kept
CLS, DLS (this thesis)	gather at a hard, projected index	discarded
token indicator (ours)	$\log p(v \mid \mathbf{x}_{<i})$, indicator relaxed	kept

- No, they do not share the blindness. Both relax the *target* as well as the input, which is exactly the design that makes the self term differentiable. MuCoLa says so explicitly: $\tilde{\mathbf{e}}_{n+1}$ receives gradients “directly from $-\log P$ ” as well as through the later hidden states.
- The blind object is mine. This implementation scores the masked position by gathering at the projected token id, so $\nabla_{\mathbf{e}_i}$ reaches the suffix only. The null is about *that* energy in embedding coordinates.
- And the repair converges with theirs. MuCoLa’s self-gradient is $h_n^\top (\mathbf{e}(v) - \mathbf{e}(x_i))$, the logit difference. The token-indicator self term is $\log p(v \mid \mathbf{x}_{<i}) - \log p(x_i \mid \mathbf{x}_{<i})$. Those agree up to the shared denominator, which cancels. The term worth $0 \rightarrow 40\%$ is the one MuCoLa already had.
- What was reimplemented faithfully is their sampler *geometry*: continuous embedding state, nearest-neighbour projection, plus the correction they omit. Not their energy.

Post-submission: crossing the bridge to controllable generation

The thesis diagnoses the null and demonstrates the repair on *one-token recovery*. It never crosses to generation. Four questions were posed after submission, in priority order.

- **Q1 Cost.** The output-side surrogate splits into a *self* term $\log p(v \mid \mathbf{x}_{<i})$, already computed by the forward pass, and a *future* term costing a backward pass. Is the future term worth its cost?
- **Q2 Conditioning.** Do bidirectional proposals beat the autoregressive output-side proposal when the attribute is global?
- **Q3 Mechanism.** How should a constraint enter: exact rescoreing on a shortlist, a first-order surrogate, both, or the accept step only?
- **Q4 Usefulness.** At matched compute, does the exact-energy chain beat MuCoLa on adherence versus fluency?

Q1 and Q3 are answered below. Q2 and Q4 were gated on Q3 and were in flight; they are reported as not done rather than omitted.

Post-submission: the repair needs *no* gradient at all

Same exact-energy Metropolis–Hastings chain, full vocabulary, the cell that produced the headline one-hot result, $n = 200$.

Arm	Exact %	Final KL	fwd-eq	bwd passes	GPU s/acc.
self term only	24.5	4.598	100	0	0.332
self + linearized future	26.5	4.584	400	10,000	0.881
input-embedding gradient	0.0	6.243	300	10,000	1.391
uniform	0.0	6.671	100	0	0.857

- Paired: one-hot minus self = +2.0 exact points $[-0.5, +5.0]$, $\Delta\text{KL } -0.015$ $[-0.361, +0.331]$, for four times the compute.
- The self term *never calls autograd once* (the run's own counter certifies it) and still recovers 24.5% where the input-embedding gradient recovers 0.0% on the same sequences.
- Section 5.6.1 leaves open whether self and future “come apart in the interior”. Measured in the interior: not to any degree the data can resolve.

Post-submission: the 40% headline at $n = 200$

n	Exact recovery	95% CI
50 (the sweep cell as published)	40.0%	[26.4, 53.6]
200 (same configuration, rerun)	26.5%	[20.4, 32.6]
input-embedding gradient, $n = 200$	0.0%	—

- The published number **reproduces exactly**: the same first fifty sequences give 40.0%. Nothing is wrong with the code or the data; 40% was an $n = 50$ estimate.
- What does not survive: the parity claim. SEDD-small 38.5%, SEDD-medium 39.0%, RoBERTa-large 44.5% were all measured at $n = 200$. At equal n this sampler is *below* them, not comparable to them.
- What does survive, and it is the load-bearing part: 26.5% against 0.0% on the same sequences, paired gap 24.5 points, CI [18.5, 30.5], now on four times the data. Direction, mechanism and diagnosis untouched.

Post-submission: the floor nobody measured (shortlist width)

All arms proposing over the *same* frozen top- k candidate set, $n = 200$ per cell.

Arm	Full vocab	$k = 256$	$k = 64$	$k = 16$
uniform (no scoring at all)	0.0	8.0	18.5	31.5
input-embedding gradient	0.0	8.5	21.5	30.5
self term	24.5	26.5	29.5	30.0
gradient minus uniform (pts)	0.0	+0.5	+3.0	-1.0

- At $k = 16$ every arm lands between 30.0 and 31.5 percent, *including the one that scores nothing*: the support and the exact accept step are doing all the work.
- This does **not** rehabilitate the input-embedding gradient. Its correct control is the uniform arm, and it tracks that floor at every width.
- It **strengthens** the null: holding the support fixed rules out the objection that the null was an artifact of proposing over 50,257 candidates. This control was never run in the thesis, and I would add it if I did this again.

Post-submission: a pre-registered prediction that was refuted

Exact future term (energy differences on the shortlist) against the self term.
Prediction recorded in advance: “exact scoring buys little for its cost”. Wrong, and reported as wrong.

k	$\Delta\text{KL} (\text{exact}_k - \text{self})$	$\Delta\text{exact} (\text{pts})$	Cost, fwd-eq	Cost, wall
16	+0.090 [−0.264, +0.442]	+0.5 [−3.0, +4.0]	×1.17	×1.04
64	−0.555 [−0.977, −0.161]	+1.5 [−2.0, +5.5]	×1.65	×0.86
256	−0.601 [−1.008, −0.205]	+4.5 [+0.0, +9.0]	×3.57	×1.32

- Once the shortlist is wide enough for ranking to matter it buys about 0.6 nats, and at $k = 64$ it does so at 0.86 times the *wall-clock*: faster than the arm it beats.
- Why: acceptance 66–80% against 23–30% for every other arm, so far fewer wasted steps.
- Caveat I would state unprompted: with one masked position the k candidate energies do not depend on the current token, so they are computed once per sequence. That economy is task-specific and will not survive to a task where many positions move.

Post-submission: why constraint gradients are weak here

At each step: find the token the steering classifier most prefers over the whole vocabulary, then ask where it sits in the likelihood ranking. $n = 30$ per cell.

k	Coverage	Chance	Median likelihood rank	Linearization ρ
16	0.2%	0.032%	29,928/50,257	+0.019
64	0.8%	0.127%	34,227/50,257	+0.040
256	1.7%	0.509%	29,866/50,257	-0.008

- The shortlist almost never contains what the constraint wants. The two rankings are close to independent. Coverage is 3 to 6 times chance, so the association is real, but under 2% even at $k = 256$: a weak exact-rescoring arm is then a *support* limitation, not a verdict on rescoring. Pre-registered.
- The constraint-side linearization is in the same condition as the input-embedding gradient. Spearman against the true change: token-indicator surrogate +0.60 to +0.73, input-embedding gradient ~ 0.0 , classifier gradient +0.02 to +0.06.
- A qualification of concern 11, not a contradiction: a classifier gradient can point usefully in aggregate over many small continuous steps while poorly predicting any single discrete substitution.

Post-submission: Q1 on the generation task

Constrained continuation, $n = 60$, span 20, 600 steps, $k = 64$, constraint in the accept step only. Partial: `exact_k` and `uniform` still running.

Arm	Steer-head acc.	Held-out adherence	Accept %	fwd-eq/seq	GPU s
self	36.7%	46.7% $\pm$ 12.6	10.3	635	2,982
onehot	41.7%	63.3% $\pm$ 12.2	10.3	4,133	8,047
embgrad	38.3%	56.7% $\pm$ 12.5	12.9	4,234	10,712

- The apparent ordering is **not** a separation: ± 12.5 point intervals that overlap heavily. The better powered paired test puts both differences on zero: onehot minus self $+0.138 [-0.186, +0.449]$, embgrad minus self $+0.034 [-0.255, +0.320]$.
- So Q1's answer does not change when the task changes from repairing one token to conjuring twenty: the self term matches the full surrogate at $6.5\times$ less compute.
- **One result the other way:** embgrad reaches $+12.52$ nats higher final LM log-probability, CI excluding zero. Not directional signal: all arms share the identical exact accept step and target the identical distribution, so this is *mixing*.

Post-submission: three defects caught before any real run

Each would have produced a confident and wrong answer. All three were found by smoke tests.

- ❶ Shortlist excludes the incumbent. The exact arm put zero mass on the token the chain currently held, so the reverse probability was zero, $\log \alpha$ was $-\infty$, acceptance 0.0% on every sequence.
- ❷ The distance term pins a shortlisted chain. With $k = 64$ and t_1 present, all five arms returned an identical final KL of 6.777: t_1 scores the incumbent at 0 and every rival negatively. Over 50k candidates that is one preference among many; over 65 it is decisive.
- ❸ A state-dependent shortlist is not reversible. Recomputing “top- k here plus where I am” gives the reverse move probability zero as soon as the chain moves. Acceptance 0.0% while the proposal was healthy: 24.6% of proposal mass on the ground truth, the move worth +19.8 nats, refused because the reverse term was $-\infty$. Fixed by freezing the support per sequence.

Regression gate after every sampler edit: the flagship archived configuration reproduces bit-identically, maximum absolute difference 0.000e+00 on all three recorded channels.

Post-submission: what was *not* done

Stated rather than omitted.

- Study B (proposal sources, RoBERTa and SEDD on the constrained task) was gated on the Study C mechanism probe and was not launched. Both proposals are supported by the harness and were validated in the recovery setting.
- Study C's full mechanism comparison at scale was in flight; only the coverage half above is complete.
- Study D's adherence-fluency curve at matched compute was in flight.
- Study E (diversity at matched adherence against a compute-matched best-of- N optimizer) was not reached. The metrics are implemented and validated (distinct-1 0.929, distinct-3 1.00, self-BLEU 0.000, semantic spread 0.778 on one arm) but the *controlled* comparison is the part that matters, so no claim about posterior coverage is made on this evidence.

Two harness lessons that generalize: a single-mask diagnostic task hides a class of scaling bug (two defects here were invisible at one masked position, immediate at twenty); and reversibility failures are silent in the worst way, acceptance collapsing to exactly zero while the proposal itself is perfectly healthy.

Post-submission: five sentences

- ❶ The constructive half of the thesis needs no gradient at all: the self term, which every autoregressive model already exposes, gets the whole effect at a quarter of the cost and zero backward passes.
- ❷ The 40 percent headline is an $n = 50$ cell that reproduces exactly; at $n = 200$ it is 26.5 percent, below SEDD and RoBERTa rather than at parity. The zero-versus-nonzero contrast the thesis rests on is unaffected and better supported.
- ❸ The scope restriction the thesis states about MuCoLa is correct and can now be demonstrated rather than argued: the same code with MuCoLa's energy writes fluent English, with mine it writes word salad.
- ❹ A shortlisted sampler needs a uniform-on-the-same-shortlist floor before any surrogate claim can be read, and that floor was never measured in the thesis.
- ❺ Gradient-based constraint guidance is weak here for a measurable reason: the classifier's first-order expansion has rank correlation near zero with the change it is expanding, and a likelihood-ranked shortlist almost never contains the token the constraint wants.